

Ellison Hall Case Study - *The University of California, Santa Barbara*

Building Description

Ellison Hall, built in 1969, is a large, multifunctional office building on the University of California, Santa Barbara campus, serving various academic and administrative purposes (see Figure 1). Located near the northeastern edge of the main campus, the building operates 250 days a year. It supports approximately 600 faculty, staff, and graduate students across various departments, including Political Science, Sociology, and Geography. It contains approximately 300 high-occupancy offices, 12 classrooms, and meeting rooms, and is actively used for instruction and research. With a gross floor area of 118,172 square feet, it plays a vital role in campus operations and community engagement. According to the U.S. Green Building Council¹, Ellison Hall was certified by LEED on July 26, 2012, and achieved a Gold certification (50/85) under the LEED O+M Existing Buildings v2.



Figure 1. Ellison Hall

Energy Assessment

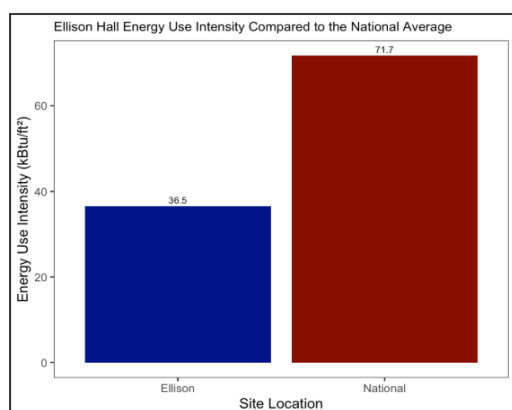


Figure 2. Ellison Hall EUI compared to National Average

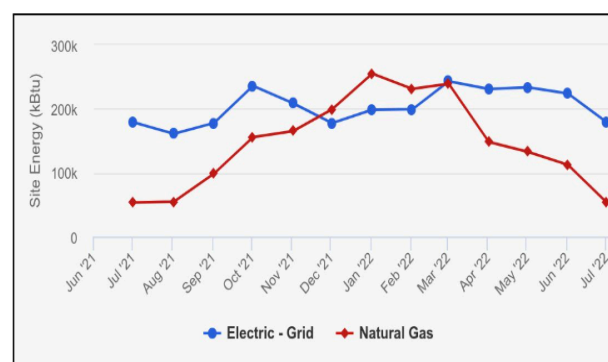


Figure 3. Energy use by calendar month

After benchmarking the building through the ENERGY STAR Portfolio Manager², Ellison Hall achieved a strong ENERGY STAR score of 88 for the year ending July 31, 2022. This places the building well above average energy efficiency relative to similar office buildings across the U.S.

¹ U.S. Green Building Council, *Ellison Hall*, <https://www.usgbc.org/projects/ellison-hall?view=scorecard>.

² U.S. Environmental Protection Agency, *Benchmark your building using ENERGY STAR® Portfolio Manager®*. <https://www.energystar.gov/buildings/benchmark?testEnv=false>

The site energy use intensity (EUI) was 36.5 kBtu/ft², significantly better than the national median of 71.7 kBtu/ft². The source EUI was 49% lower than the national median (see Figure 2). Approximately 57% of energy use came from grid electricity and 43% from natural gas, resulting in 262 metric tons of CO₂ emissions annually. However, monthly data (Figure 3) show a sharp increase in natural gas use during the winter, likely for heating. In colder months, gas use even surpasses electricity, revealing a seasonal dependence that drives up emissions. Reducing this reliance through improved insulation, electric heat pumps, or integrating solar photovoltaic (PV) systems would further lower Ellison Hall's carbon footprint. Despite lacking onsite or offsite green power, the building's performance is commendable given its age.

Water Assessment

According to the Indoor Water Use Calculator, Ellison Hall's design water consumption is **872,400 gallons annually**, exceeding the baseline of **732,575 gallons** by nearly 140,000 gallons (see Figure 4). This reflects a **-19.09% efficiency change**, indicating that current retrofits have not significantly improved overall water use. The building is equipped with dual-flush toilets installed across all restrooms. Male restrooms had one low-flow urinal (0.125 gallons per flush (gpf)), and the rest were outdated conventional urinals (2.0 gpf), contributing to higher water usage. According to the Water Use Calculator, flush fixtures account for **approximately 471,175 gallons of water per year**.

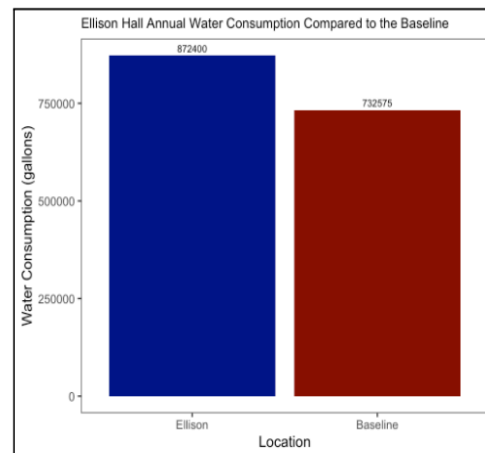


Figure 4. Ellison Hall's annual water consumption

Flow fixtures are another key issue. Public lavatory faucets operate at **2.2 gallons per minute (gpm)**, well above the recommended **0.5 gpm** for efficient buildings. With approximately 50 such sinks, this results in a design flow volume of **401,225 gallons per year**, exceeding the baseline. While some improvements have been made, outdated urinals and high-flow faucets are the primary contributors to inefficiency.

Current Sustainable Features Utilized

Despite its age, Ellison Hall showcases various sustainable design features and operational strategies that reduce environmental impact and enhance occupant comfort and efficiency.

1. Dual-flush toilets

One of the most noticeable features is the installation of **dual-flush toilets**, which enable users to select a lower or higher volume flush depending on the type of waste. This small user-controlled behavior change has a substantial impact on overall indoor water consumption, particularly in buildings with high occupancy and frequent restroom use.

2. Motion-sensor lighting

Another major component of Ellison Hall's sustainable features is the **motion-sensor lighting** in stairwells. These sensors help conserve electricity by reducing energy use during periods of inactivity. They ensure lights are only activated when needed, preventing waste during nighttime or low-traffic hours.

3. Overhangs on the windows

Ellison Hall also incorporates **architectural overhangs** above its windows, which passively reduce solar heat gain during the warmer months (see Figure 5). This design strategy is particularly effective in Santa Barbara's coastal climate, where seasonal sunshine is abundant. By minimizing direct sunlight exposure, these overhangs reduce the need for excessive air conditioning, lowering energy demand and improving thermal comfort for occupants. Although its purpose is to reduce the glare of sunlight during peak hours, it still allows enough light into the building so that occupants do not need to rely on bright lighting, which can lead to *sick building syndrome*³ if exposed for long periods.



Figure 5. Window overhangs on Ellison Hall

4. Bike rack near the promote alternative transportation

The building is also embedded within UCSB's walkable and bike-friendly campus. A **bike rack is located adjacent to the building**, encouraging students and faculty to cycle rather than drive. The bike rack is closer than the nearest parking garage, where most students and faculty can park. This supports a mode shift toward alternative, low-carbon transportation. The location also benefits from excellent pedestrian connectivity and easy access to public transit, reducing reliance on single-occupancy vehicles.

5. Multi-purpose building

From a spatial efficiency standpoint, Ellison Hall's interior layout supports sustainability through **multi-functionality**. The building features offices, classrooms, and conference rooms within a single footprint, thereby reducing the need for separate structures. Multiple graduate students or staff share many offices, promoting better use of conditioned space and minimizing per-capita energy consumption.

³ Joshi, S. M. "The sick building syndrome," *Indian Journal of Occupational and Environmental Medicine* 12, no. 2 (2008): 61–64, <https://doi.org/10.4103/0019-5278.43262>.

These features reflect an intentional effort to embed sustainability into Ellison Hall's daily operations and physical design, despite its original construction date of 1969. The combination of passive design elements, efficient space utilization, and behavioral nudges, such as dual-flush toilets and motion-sensor lighting, all contribute to a lower environmental footprint. However, while these efforts are commendable, they also reveal a broader truth: even buildings that appear environmentally conscious still hold untapped potential. Ellison Hall offers a strong foundation, but with targeted investments and improvements, it could transition from a relatively efficient structure to a high-performance building aligned with UCSB's long-term climate action plan⁴. In this way, the building serves both as a success story and a reminder of the ongoing opportunities for improvement that exist even in well-performing spaces.

Opportunities for Improvement

Although Ellison Hall performs relatively well in terms of energy use and has several incorporated sustainable features, it still presents meaningful opportunities to enhance efficiency, occupant health, and environmental performance. According to the technical assessments, the most pressing area of concern is **water consumption**, which exceeds baseline estimates by nearly 20%. Fortunately, many needed improvements regarding water consumption are low-cost, high-impact interventions that could be implemented with minimal planning. Along with other improvements related to air quality and reducing thermal loss, these upgrades would support UCSB's broader climate goals and enhance the daily experience and well-being of building users.

Install 0.5 GPM Faucet Aerators in Public Restrooms

- The most cost-effective and high-impact upgrade is to replace the existing **2.2 gpm** faucet aerators in all restroom sinks with **0.5 gpm** high-efficiency models⁵. During the building walkthrough, approximately 50 sinks were identified with outdated fixtures. Replacing all of them would cost roughly \$110 plus tax, making it an extremely affordable retrofit. The improved efficiency would result in an estimated **annual water savings of 107,312.5 gallons**, increasing the building's water efficiency from -19.09% to **+14.65%**. In addition to reducing water waste in daily use, these lower-flow aerators minimize water loss in the event of a leak or an unattended faucet.

⁴ University of California, Santa Barbara, *Climate action*, UCSB Sustainability, <https://www.sustainability.ucsb.edu/focus-areas/climate-action>.

⁵ NEOPERL, *0.5 gpm Flow Rate, Spray Flow Outlet, Outlet - 10N182*, Grainger Industrial Supply, <https://www.grainger.com/product/10N182>.

Replace Urinals with Waterless Fixtures

- Another impactful upgrade would be to **replace the existing porcelain urinals**, many of which appear original to the building and are inefficient by modern standards, with waterless urinals⁶. These outdated models consume a substantial amount of water with each flush, and based on visual assessment, these units can hold 1.5 to 2 gallons in the bowl (see Figure 6). Swapping them for waterless versions could yield an additional **annual savings of 75,862.5 gallons**, or **10.36%** above baseline. Combined with the proposed aerator replacement, Ellison Hall's water savings would be 44.09% above the baseline, totaling over 323,000 gallons saved per year. Although straightforward, there were 17 standard urinals, and the waterless urinal costs around \$389 per unit, totalling approximately \$6,613 (plus \$628.24 in tax), not including labor.



Figure 6. Urinal fixtures in Ellison Hall

Replace Floor Mats to Improve Indoor Air Quality

- While the results of the improvements are less quantifiable, **replacing floor mats** at all entrances could significantly improve indoor air quality⁷ by reducing the amount of dust and particulate matter entering the building. The current mats are visibly worn and ineffective (see Figure 7). A total of 14 commercial mats, such as the WaterHog ECO Elite⁸, would be sufficient to cover both indoor and outdoor entrances. These mats are made from 100% PET post-consumer recycled fiber and 15% recycled rubber, supporting UCSB's materials sustainability goals. Their durability and ease of cleaning also reduce long-term maintenance costs. High-quality floor mats can capture up to 80% of incoming dirt and pollutants⁸, lessening the burden on HVAC systems and promoting a healthier indoor environment.



Figure 7. Entrance floor mats into Ellison Hall

⁶ Sloan, *Vitreous China, Waterless Urinal - 32U918*, Grainger Industrial Supply, <https://www.grainger.com/product/SLOAN-Waterless-Urinal-Sloan-32U918>.

⁷ American Floor Mats, *Waterhog ECO Elite entrance mats*, <https://www.americanfloormats.com/waterhog-eco-elite-entrance-mats/>.

⁸ Autumn Roe, "How floor mats play a crucial role in maintaining indoor air quality," Swan Dust Control, May 12, 2025, <https://www.swandust.com/blog/how-floor-mats-play-a-crucial-role-in-maintaining-indoor-air-quality>

Upgrade to Double-Pane Windows

- One capital-intensive but highly beneficial upgrade would be **replacing the original single-pane windows with double-pane units**. These single-pane windows are responsible for considerable thermal energy loss, particularly on upper floors that receive more solar exposure and experience greater air temperature fluctuations. Upgrading to double-pane windows would enhance several aspects, including **thermal insulation, blocking harmful UV radiation**, and significantly **improving occupant comfort and productivity**. In a high-traffic academic setting like Ellison Hall, these benefits not only reduce operational costs but also contribute to a healthier and more consistent indoor environment for faculty and students..

These proposed upgrades, ranging from simple fixture replacements to more capital-intensive window retrofits, show that substantial improvements are possible even in buildings that already incorporate sustainability-minded practices. The university can achieve measurable resource savings and demonstrate leadership in cost-effective building performance by addressing Ellison Hall's most immediate inefficiencies, especially in water use. Each recommendation aligns with the **triple bottom line**: reducing environmental impact, lowering operational costs, and improving health and comfort for occupants. The investment return would be quick and significant in many cases, particularly the faucet aerators and waterless urinals.

Lessons Learned

This assessment of Ellison Hall revealed just how inefficient older buildings can be, often not due to major structural flaws, but because of small, overlooked inefficiencies that accumulate over time. Water use, in particular, stood out as the most immediate and solvable issue. Replacing something as simple as faucet aerators can lead to dramatic savings with minimal cost, proving that meaningful change doesn't always require significant renovations.

One of the most important takeaways is that sustainability must be closely tied to occupant well-being. A building could be designed to meet every technical metric of green performance, but if it feels cramped, dimly lit, or disconnected from nature, it becomes a place people don't want to spend time in. Good indoor air quality, natural light, and access to green space are not just luxuries but essential to mental health and long-term productivity.

This project also showed the value of a focused, category-specific approach. Instead of trying to address every sustainability category at once, it was more effective to prioritize a few areas in detail. Identifying specific weaknesses and weighing the pros and cons of the upgrades is a much more effective approach. This method made it easier to prioritize improvements and understand how each recommendation fits within the more extensive building performance system.

Ultimately, this case study reinforced that sustainable building performance is not static; it must be continually revisited, reassessed, and refined as emerging technologies hit the market. What was once sustainable 20 years ago could easily be the new baseline that buildings must meet. Even buildings that perform well on paper can reveal clear opportunities for progress once you investigate them closely, primarily through firsthand observation and practical data analysis.